25901897 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(Honours)/MGU-BBA(Honours)/MGU-BCA(Honours)Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECO201 - QUANTITATIVE ECONOMICS - I

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Question

Answer all questions

Each question carries **1** marks

1. $A \cap \emptyset$ = ? [K] / [CO1]

   a). A b). Ø

   c). A∪Ø d). None

2. If P(A)=0.5, P(B)=0.5, P(A∩B)=0.2, then P(A∪B)= [U] / [CO2]

   a). 0.7 b). 0.9

   c). 0.05 d). 0.2

3. Linear consumption function is [U] / [CO1]

   a). Qs=f(x) b). Qd=a-bP

   c). Qd=f(y) d). C=a+bY

4. The condition for profit maximisation is: [K] / [CO2]

   a). MC=AC b). MR=MC

   c). TR=TC d). R=AC

5. Parameters are: [K] / [CO1]

a). Always fixed values. b). Values controlling a model/equation

c). Always variables. d). None

6. If f(x)=x^2 and g(x)=3x, then according to the product rule: [U] / [CO2]

a). f ' g + f g' b). f' g'

c). f' g d). f g

7. The demand function is [U] / [CO1]

a). Qd=f(p) b). Qs=f(x)

c). Qd=f(y) d). Qd=f(I)

8. What does the derivative represent in calculus? [U] / [CO1]

a). Average Value b). Area under the curve

c). Slope of the tangent line d). Accumulated sum

9. A matrix with equal number of rows and columns is a: [K] / [CO1]

a). Diagonal matrix b). Singular matrix

c). Equal matrix d). Square matrix

10. From the function $y = 6x^2 - 12x$, the second - order derivative is: [A] / [CO3]

a). $\frac{dy}{dx} = 6x - 6$ b). $\frac{d^2y}{dx^2} = 12$

c). $\frac{d^2y}{dx^2} = 6$ d). $\frac{d^2x}{dy^2} = 12$

[1x10 = 10]

Part B

Short Answer

Answer any **10** questions

Each question carries **3** marks

11. Explain demand function. [U] / [CO1]

12. Draw a total revenue curve from the function P=80-2x [U] / [CO1]

13. Find A - B and B - A . [U] / [CO1]

$$A = \begin{bmatrix} 5 & 2 \\ 3 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 6 \\ 4 & 2 \end{bmatrix}.$$

14. Find the minimum value of f(x). [U] / [CO2]

$f(x) = x^2 - 4x + 5$

15. What is a random experiment? Give an example. [K] / [CO2]

16. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, verify that $(A^T)^T = A$. [A] / [CO4]

17. Differentiate the following, [An] / [CO2]

i. $x^2 + y^3 = 64$

ii. $5x^4 - 4y^3 = 200$

18. Explain the relation between P(A∩B) and conditional probability. [K] / [CO2]

19. Solve the system of equations using Cramer's rule: [A] / [CO5]

2x+3y-5 = 0

-3x-2y-1 = 0

20. Compute A×B and B×A if it exists [A] / [CO4]

$A = \begin{bmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 3 & 1 & 5 & -1 \\ 4 & -2 & 1 & 0 \end{bmatrix}$

21. Find f'(x). [U] / [CO3]

$f(x) = (2x + 3)^5$

22. For the matrix [U] / [CO1]

$D = \begin{bmatrix} 3 & 2 & 0 \\ 0 & 0 & 1 \\ 2 & -2 & 1 \end{bmatrix}$

(a) Find the minor of M_{13}

(b) Find the cofactor of C_{22}

23. Find the value of: [U] / [CO1]

1. $\ln_5 (25) = ?$
2. $\log_{10} (100000) = ?$
3. $\ln_3 (27) = ?$

24. Give the sample space for rolling a die. [K] / [CO2]

25. If A = { 3,4 } B = { p, q } find A x B and B x A [U] / [CO1]

[3x10 = 30]

Part C

Short Essay

Answer any **5** questions

Each question carries **6** marks

26. Find $lim_{x \to 3} \frac{x^3 - 27}{x - 3}$ [U] / [CO1]

27. Given the saving function S = -50 + 0.2Y and investment I = 80, find the equilibrium level of income. [U] / [CO5]

28. Find the inverse of the matrix. [U] / [CO4]

A= $\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$

29. Differentiate the following. [A] / [CO1]

1. $y = (x^2 + 3x)(x^3 + 2)$
2. $y = \frac{x^2+1}{x}$
3. $y = \frac{3x-5}{x+2}$
4. $y = (2x^2 + 1)^{-4}$
5. $y = \log(x^2 + 4)$
6. $y = 7e^{3x}$

30. Check if f(x) is continuous at x=2 [U] / [CO1]

$$f(x) = \begin{cases} x^2 + 1, & \text{if } x < 2 \\ 3x - 1, & \text{if } x \geq 2 \end{cases}$$

31. Elaborate on the role played by probability in economics. [U] / [CO2]

32. Find adj(A). [U] / [CO1]

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 4 & 0 \\ 3 & 5 & 6 \end{bmatrix}$$

33. Find the derivative of the function [U] / [CO1]

$a. y = x^6 + 4x^3 - 7x + 2$

$b. y = x^5 - 3x^4 + 2x^2 - x + 1$

$c. y = x^9 - 4x^7 + 5x^3 - x + 6$

34. Explain independent and dependent events with examples. [U] / [CO2]

35. Find AB and A+B if it exist. [U] / [CO1]

$A = \begin{bmatrix} 2 & 6 & -3 \\ 5 & 8 & 0 \\ -2 & 4 & 3 \end{bmatrix}$ B = $\begin{bmatrix} 0 & -6 & 9 \\ 7 & -4 & 3 \\ 1 & -5 & -6 \end{bmatrix}$

[6x5 = 30]